

A Packing of 17 Four-Spaces in $\mathbb{R}^{13}$

Limin Ge*

October 9, 2026

Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 2.941950. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.931715, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.941950 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 17$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.931715, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 17/16 = 2.942308$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N17.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

294195093827062395355696659381941204668560538838627685142262238172685568
531720842195286348234690319856283132337411249059546857538846569671054849
8314076209147836156338262958546674740159417107911

denominator:

999999999999110261209134097160517505999518208498034126080772038941038940
457323281332952273797816340006387929455587289968244885099418453446182545
870317074937144899423621201327642954276764155745

B Coordinates

17 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N17.txt`.

```
502634867969 451076169230 -146751012279 -81924212754766338200 79273190138 -232966033439 93028962928 123453393864 209237142270 96689008373 67592705169
-161781616556 254265624265 163036482830 -2574478658361687577279 35867851407 353280590552 -357142963933060894994 -203359799271 143780295421 192408521825
-224608302127 478999626205 -22783803195 81231074204254515859596 57255673798 -635759803848 178395257933286302316044 197878643521 367565670328 -230275103745
-356339633784 -2739502380 476028804668 108070421721-319775130179 278729260675 237652756099 5876228619743777217664 165784301737 -547593174445 -54397509023
-89431318348 -130907881660 -326698375593 -1042944822366014133 -230696852458 -93475358755 180805409780-321224427668 -514753322920 -302546030430 -32346703960
346350269721 -193071175618 -160648779156 -198657840976463033708 -155666330643 95295740567 365419930535592850385 25413100648 -91063021318 -699445781977
78450888436 246113392248 438078638738 -70609485547 200299598076 -82985450361 92337112279 243858579091 -139516298311 262187253173 330728255752 9071905854
-62858517130 216125670262 -274953698239 16136456333676310829767 -110405411329 387789091847 -1650934492435958931926 303745024281 -311002691334 -155378557336
415988489016 88203699885 376905425219 -524844595852-11511012073 652557874389 270648314554 38522253070724144978604 -6494880145 -36725976660 -404844645693
-139236965674 -10889990806 10058071344 -1627887933727273407534 -165662014043 306973047842 2939304147870524822992 258595466485 -92137152030 -231821655273
-440062838052 20335385401 -213336523400 -7061970513679096003 325051221314 -95410743427 -489756868423-474584921933 183295297377 -12831447543 318235238478
-140010918345 414117514715 75693894625 149549210812136356803016 -403304135974 246266868863 3549499571289081833258 222640350851 -137565690510 12558277186
-2038296421 401161248254 -362347911800 -1047281995123649299684 261655651565 6853085675 304521712978 131309063442 -61936968406 172447778945 -293268906574
157307586550 -85438669503 -463997175319 -312866520971832878358 176533125423 22748912673 -12481669800275681089585 211835320677 -443594632348 125053472082
670258955891 -309541917557 75839494461 74420384751 6504175869 -185960634497 229374466515 -329200031617-171420762113 -554931997030 -60541837095 -93818516786
-33989434763 -286829842261 387402937018 15670415377252427961557 -82325199386 309977034298 -2315130547800120342644 114334248454 -264445374454 -264445374454 103271752055
-2860273094985 48119886418 16817431135 -74477121113 -223063783923 -27913565420 -343625905043 11829242250619073226744 5507671675 110244299466 21935865167
-271179830044 -329219913977 -173323198435 67014252286092397473 -85884397812 25399711931 -82526984226 25295518709 143612593382 -488883792098 465687858937
-46880396663 94347281649 -95403836305 169314561693 -141404725144 206341706029 -53936450280 5791409749152992944808 152700538363 -161071455002 113628341150
28585873068 -58919054373 -85982026826 401436596895 94909669667 -189553007508 326749284375 54115435025149542859966 100132442606 416775060802 -57203627692
-235659388763 482648234293 -462846873048 325038592091456801413 10360147070 -126894213932 54571307610 325593125461 -68913011578 76093880322 -17234968215
107471256237 232803915867 -13006599106 22318369800518148628422 108559959939 -276718574694 -58903291825606029180 -228140089753 -187128843178 -260149885353
290644544418 -1385953460 -254302549973 106321064599-260828704387 -15742042088 382396021209 -2374062544116284853047 135813615201 333745604829 17819963373
166029122601 71941304571 271006762630 642095460168 -61925424843 -759010431300 -519684732473 -45657227306525894022 -572650493706 141576379212 649060760357
87435618345 -236533424438 -402320665709 207031208725151919749470 -26402601005 75841570815 -1074712910187816668366 364720976227 163369132439 147875187723
257521494969 58588908729 260158853127 -2241659262268551889373 -5052572624226 326981120567 33454932668400056290860 -184483014390 -493376601786 -208515582171
141308075018 231735309149 113244646856 247230263408-164030993065 77305289199 -355049635445 -2461050277907028633283 460929636848 -187238867848 -89973604953
-423993480856 -72811371973 42674507184 34090754801518221309936 97687146805 -472374452263 -1992086716366459523232 -387616401576 -43083525081 -404207941404
-18890896305 151681535834 558589864866 -714421370207856614680 374660173136 -321105835443 -2209851407527137721997 -428669864005 -164202795336 -70258732203
-132630112100 31190387591 2547779919829 288527800023-306715744001 -291991279371 -265846240079 18166093387698090034 -286461265837 46585984595 -137799499511
-228938390932 48144670795 -430886340733 453366047682087464089 -326688547553 -167990391425 -3359056421211090567607 -83643947461 102928462037 -132510250693
569292161576 -149583891127 137845297622 47579911103795570351726 296576108688 -37656339396 -3823515822388995278464 274665130836 -416498749695 -253105739883
-36551630362 -54679712056 486306372192 3218748177095191915 340676695516 -277827202441 98236995118 -65772730564 -81805328455 -33083525081 -404207941404
-65981027620 -616022584862 28860542088 -2695108181082427315405 -58831897158 156802248294 -291653220238121805909 198901873722 -149430487439 268972796082
52438482098 35487446850 -143153762385 -32819703935160034938670 -56261591921 51880216141 -69109172436 168546267033 -606452655653 -178340265156 370458793301
32565158910 370406535674 252076130235 10381762806 -3501040084 -598636377385 -185705978727 25473095433-32139943478 138647925931 101679893556 67657541812
-27696645847 495343518974 -179455667983 -239111083032696001646 -183657305814 226316144715 -307393110136363646230 137979699205 -54921627740 -516464463259
-216204309239 29512649575 -214265258927 241111230270624487564 -149835339489 -113926329234 41517452388270664623428 2219313088 527130540226 229383142110
276066214609 -353770571704 65440447017 74695882661 307272598101 769363071303 -53058886452 485756294508 305152530012 382876668696 175212061175 364158824824
18406070340 -342897750350 66787456892 83799178969 201760888556 -164294738522 -322562746720 -22242445643841556886 -52136279550 37826058170 -343312708699
-94153591175 144253884126 -139200427187 2639179205826579134699 -151831387440 -245578748793 9161072044637676693237 6115144636 -304576716575 290988916316
16832065152 50585178732 330898518800 -460119270867 -352618495854 -107893787143 -230019728065 -26742208533186545097 42535935300 9123830662 -8380173565
-176829060514 -179340964601 194552480637 -314912124450275327832 280273601808 32514210002 -186459148507-230472377875 -58966214627 247141691771 -18361133929
290008670907 119512809463 752341129787 136344304423164836668118 -433150876872 583319405954 -451997769921253002595 173354382685 -154271828681 344036966631
-589355146029 81651619735 153323332161 -46918050393-743046314855 -57214311988 344731886961 11349832195151053356449 -239201513315 16814582385 -622320548259
-336322539044 -3956550754631 377485520286 167929417443619766821 -34233652934 -322618115156 -4035885829057133923807 -192777352255 234225806185 -46343074824
469432915280 -1601341800064 -60795085449 -681060389373754086397 86459072687 -139813790069 55919086230 -150146229664 -90846801498 -18565948248 -380144562306 57286023765
281941737308 32821757410 25549811859 415113398389 -227355275927 9145151335 13507281525 -45821485674674956623113 -1902002401531 290546589751 -29990790123
91861023138 298738767190 96737251844 -1144001873328 -106171991392 912518488552 -145367946152 -74213780359354344312 288314987210 -36476201734 206331927167
9065609173 64197873592 -254723203612 -254543287681 -306556976784 139987135759 -269590219885 163642046220179340837 16830361483 400403673795 95361062723
92373622659 -162555985396 714797258084 -52212666397758628797750 -276399637853 -55007298340 190526456636399601309 -39725189568 191251862637 137646120692
312690247646 408558680287 -121709154514 -18997827623082795232793 1879399543 360041835637 459823170452 39470481807 145097503537 -511311854516 -46370411923
-5112886816 -217642182098 486722676393 -4060355132510474566591 -757825510256 -276303785924 132661291445228229111341 -301359065535 -133282351273 382900712808
446265665201 219840875321 -217218707426 166070642255047273643 161127251037 193389931546 -25503751949486265667543 -679586958248 -380144562306 57286023765
-242242953914 -116619288976 226010201839 -490671482181448269803 353301081493 -187254765015 23038571056000595131647 -122442147747 660350172642 -68257377680
-2138827658 -65222375224 -331703731304 -2470898922032616936694 71649720585 -11825840439 -29068696923040913201 -339832083777 50211767697 -198470709511
-5441830939 -296228727203 -15771365730 -23303866539396360801 113349284783 107806747500 157042943449-26476016229 364643356574 371972244601 -281952407796
-653921763696 165067541623 252510231381 -154423265326494699731 -205252082136 -46486524498 -280230933157684947809 -103104702437 112251386699 189260055992
-81563525036 126086346568 -33888663057 71912511773 -252914549280 443126836336 -240132759270 19838267233045532456695 -205042495122 -270837045900 204820795415
189905956822 449107777875 151856865134 264673264818-72117096055 -605863041266 -4724093573919 126240178428098677119 641712627474 -21631962987 -192690027124
24460983140 -32521466116 -222116115410 -30125818236777479643 -17198856000 -336322888230 15322516541 -408811924780 141739375764 -230909048382 -77717247916
88502396583 626201127753 -258009452436 -5416657858375544523803 176830036064 -415753331016 40395128644295507470367 -243677421554 26689423579 237075834213
-472998782762 -146711578887 -344622764770 34807733339328688218 158576692464 63773793398 -3421257523752409278629 -247121013771 -33639981817 221422984715
169997579688 -159416866525 -391286119783 40156280361961287937 249830926731 -222049530351 -57673321058819869697079 -184139167339 100612267071 458110988372
```

10391472149 118765895839 -13322691042 369181863491 244334754015 -478590800270 -234868594635 4884573702271101094520 30127882307 -36929264572 -196317791865
-273667056939 13746813368 -519610488190 -30052950471269001006718 128839300234 -7351071940 17544854871345061823175 -228380090165 -271246543293 82845004892
-277210413999 -288062885898 34260362136 -46238599109507963106571 106217209759 89136204726 -37430331277253678584600 606626739143 -284619526568 439625642127
365928047839 226574251171 -548326113378 -2176782751338427487044 -1211835781 64938107422 108957072876 401580700788 -159548231919 292467079343 -233149015780
41555934621 -1025702374 -55461797020 202309580318 106472434425 354565952579 11394193837 444489446865 115087484806 162983782452 -185809697032 183723155286
-284518228393 522011365112 164074521239 36383955745577031807020 -102202844829 112536303213 270351622219396793975 -438374835384 -566348954476 296440015
-133883779187 99832265467 -598204616112 -41375700552332398755 -170164200760 330100295321 28386680925292023862640 -110690164409 126932636653 -78803306315
258260306154 149650631687 283898199936 50928437020 398825498303 413013435444 -45987525478 -31991063831370515521723 -129211088679 321953133279 614493292543
-314134636952 -499471994478 140853046744 2544102191252186995463 248356402545 113941362036 -29949483849558567726962 69163904938 -257707000090 122982704294
415904983167 35190985664 361775786241 -286960975259177327756394 -199496057828 -304955061271 -40688105200